

LECTURE 1.3



Estimators and Sampling Variation

ECONXXXXX Econometrics I

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PLACEHOLDER TERM

Where We Are

1. Foundations

1.1 Data and Variation

1.2 Conditional Means

1.3 Sampling Variation

2. Simple Regression

2.1 Fitting a Line

2.2 Goodness of Fit

2.3 Bias and Precision

2.4 Reporting an Estimate

3. Multiple Regression

3.1 Adding Controls

3.2 Functional Form

3.3 Joint Tests

4. Limits

4.1 Diagnosing Violations

4.2 Limits of Regression

Last Week's Lecture

Lecture 1.2 defined the average we cannot compute, and its algebra:

1 **Expectation.**

A property of a distribution, not of a sample.

2 **The Algebra of Expectations.**

The rules every derivation in this module uses.

3 **The Conditional Expectation Function.**

The best guess of earnings given schooling.

Lecture Plan

1 What Makes an Estimator Good.

Unbiased, consistent and efficient are three separate properties.

2 The Sampling Distribution.

How an estimate varies across samples nobody drew.

3 Confidence Intervals and Tests.

Two claims about the procedure, not about the number.

Reading: Wooldridge (2020), Math Refresher C, sections C-2, C-3, C-5 and C-6.

SECTION 1

What Makes an Estimator Good

1. What Makes an Estimator Good
2. The Sampling Distribution
3. Confidence Intervals and Tests

An Estimator Is a Rule, Fixed Before the Draw

This section asks what makes one estimating rule better than another. We want average hourly earnings in the population and we hold 120 people:

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i \quad (1)$$

A Rule Before, a Number After

An estimator turns a sample into a number, by a rule fixed beforehand. An estimate is the number one sample gives.

A Hat Marks the Sample Counterpart

Every quantity in this lecture has two versions, one unknowable and one computed:

- μ : average hourly earnings over the whole population
- $\bar{y}$, also written $\hat{\mu}$: the same quantity from our 120

The Hat Marks the Sample

A bare Greek letter is a population quantity, fixed and unknown. The hat marks its sample counterpart: a rule before the draw, a number after it. So μ never changes, and $\bar{y}$ changes with every sample.



Four Candidate Estimators of the Population Mean

Each rule below could be written down before any data arrive:

- **Rule A:** Average all 120.
- **Rule B:** Average the first ten drawn, discard the rest.
- **Rule C:** Report 12.00 euro, whatever the sample says.
- **Rule D:** Drop the top and bottom five per cent, average the rest.

Rule C Is an Estimator

Rule C meets the definition: a rule, fixed in advance, returning a number.

Unbiased, Consistent and Efficient

Two rules can both be right on average, and one still lands closer. Three properties, asked separately, say which rule to prefer:

- **Unbiased:** Over the samples we might have drawn, the rule returns μ .
- **Consistent:** As n grows without bound, the rule concentrates on μ .
- **Efficient:** Of two unbiased rules, the one with smaller sampling variance.

Only in the Linear Unbiased Class

The sample mean is least variable only among linear unbiased rules. The result also needs a common mean, a common variance and uncorrelated draws.

How the Four Rules Score

No rule wins on every count, and the steadiest one is always wrong:

- **Rule A:** Unbiased, consistent, and least variable among linear unbiased rules.
- **Rule B:** Unbiased at every n , and inconsistent: ten people stay ten people.
- **Rule C:** Never varies, and is wrong by 0.48 euro every time.
- **Rule D:** Less variable than rule A, and biased downwards by the tail.

Bias and Variance Are Traded

Mean squared error is the average squared distance from μ . It equals variance plus bias squared, and that sum ranks rule D against rule A.

Why Unbiasedness Alone Is Not Enough

Unbiasedness makes two claims, and the second is weaker than students expect:

- **Across Samples:** The rule returns μ on average over repeated draws.
- **In One Sample:** Nothing, not even that 12.25 is anywhere near μ .

Unbiased and Still Worthless

Rule B is unbiased. Ten people give an answer close to worthless, so sampling variance comes next.

Choosing Among Three Imperfect Rules

You want average hourly earnings for Irish workers, and rule A is unavailable.

Question

Rule B, rule C or rule D: which would you defend, and on what ground?

SECTION 2

The Sampling Distribution

1. What Makes an Estimator Good
2. The Sampling Distribution
3. Confidence Intervals and Tests

What One Estimate Leaves Open

This section describes how our estimate varies across different samples of 120. Our sample gave 12.25, and that one number cannot separate draw from population:

- **What Varies:** The estimate, from one sample of 120 to the next.
- **What Does Not:** The population mean μ , which is one fixed number.

The Estimate Has Its Own Distribution

$\bar{y}$ varies across samples, and that variation has a distribution: the sampling distribution. Every property in this lecture belongs to it.

Five Hundred Samples, Five Hundred Means

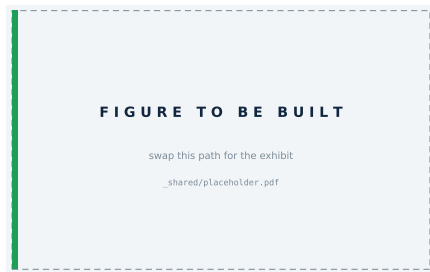


Figure 1: Means of 500 samples of 120; truth 11.52, our sample 12.25.

Our 12.25 Is One Draw

Our 12.25 is one draw from that histogram. A different 120 people lands somewhere else in it.

The Sampling Distribution Is Not Observable

In real work you draw once, compute once, and never see that histogram:

- **What We Have:** One sample, one number, and a standard error.
- **In This Module:** We wrote the population, so we can draw repeatedly.

Simulation Shows What Data Cannot

No dataset gives the sampling distribution. Simulation shows it here, and theory stands in for it everywhere else.

Spread Across People and Spread Across Samples

Two numbers describe our sample, and they measure different things:

- **5.91:** How far one person's earnings sit from another's.
- **0.54:** The typical distance between our answer and the next sample's.

Spread Across Samples Is Smaller

Dividing by $\sqrt{n}$ converts spread across people into spread across samples. Confusing the two is this section's commonest error.

The Standard Error of the Sample Mean

Errors partly cancel in a sum, so averages vary less than people do. Our 120 people are all we have to measure the spread across samples:

$$se(\bar{y}) = \frac{s}{\sqrt{n}} \quad (2)$$

- s : the standard deviation of the 120 earnings we drew
- $se(\bar{y})$: the estimated standard deviation of $\bar{y}$

Variances Add, Spreads Do Not

Variances add and standard deviations do not, hence the root. Here $5.91/\sqrt{120} = 0.54$, so four times the data halves it.

How the Standard Error Falls With Sample Size

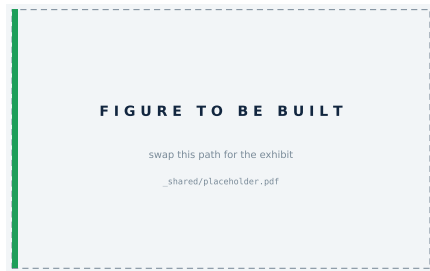


Figure 2: Means of 500 samples at $n = 30, 120$ and 480 , on one shared axis.

Four Times the Data Halves the Spread

The square root sets the rate. Four times the data halves the spread, and sixteen times halves it again.

The Central Limit Theorem, and What It Requires

Averaging washes out the shape of whatever population you started from. An extreme average needs many people extreme the same way, **which is rare**. Bell-shaped averages are what let one estimate carry a margin of error:

- **The Standardised Mean:** Subtract μ from $\bar{y}$, then divide by $\sigma/\sqrt{n}$.
- **The Large-Sample Shape:** A standardised mean is approximately standard normal.

Large Samples, Finite Variance

The theorem needs random sampling and a finite population variance, and it improves as n grows.

A Skewed Population and a Symmetric Average

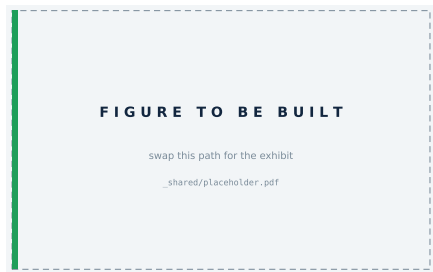


Figure 3: Left: earnings in the population. Right: means of 500 samples of 120.

The Population Stays Skewed

The population keeps its long right tail. Only the distribution of the average has gone close to symmetric.

What the Central Limit Theorem Does Not Say

The theorem is about averages, and says nothing about the people behind them:

- **Not About One Person:** Earnings stay skewed, and poor workers outnumber rich ones.
- **Not Exact:** Normality improves with n and never holds exactly at 120.
- **Two Conditions:** Random sampling, and a population variance that is finite.

Finite Samples Get No Guarantee

Nothing here says $\bar{y}$ is normal in our 120. The theorem describes a limit, and 120 buys a good approximation to it.

Doubling the Sample Size

Suppose we drew 240 people instead of 120, from the same population.

Question

Does the histogram of sample means shift, narrow, do both, or do neither?

SECTION 3

Confidence Intervals and Tests

1. What Makes an Estimator Good
2. The Sampling Distribution
3. Confidence Intervals and Tests

Building a 95 Per Cent Interval

This section turns an estimate and its standard error into a defensible statement:

$$\bar{y} \pm 1.96 \times \text{se}(\bar{y}) \quad (3)$$

- 1.96: the 97.5th percentile of the standard normal distribution

The Interval Runs 11.19 to 13.31

$12.25 \pm 1.96 \times 0.54$, and 1.96 is the large-sample value for a skewed population.

Coverage Across Repeated Intervals

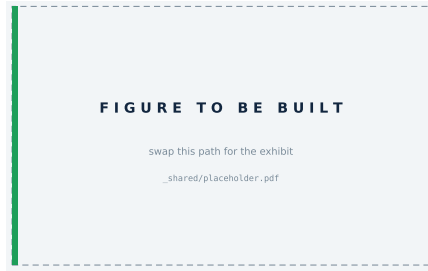


Figure 4: One hundred intervals from one hundred samples; the true mean is a vertical line.

Five in a Hundred Miss

Every interval lands somewhere different; the truth at 11.52 stays put. The rule carries the 95 per cent, not any one interval.

Why the 95 Per Cent Describes the Procedure

Throw a ring at a fixed peg. The peg never moves; the throws scatter. The endpoints jump across the hundred intervals; the truth stays put at 11.52. Randomness lives in the endpoints, never in the population mean.

Ninety-Five Intervals in a Hundred

Before the sample is drawn, the random interval covers μ with probability 0.95. Once the endpoints are numbers, μ either sits inside them or does not.

No sample says whether our own 11.19 to 13.31 covered 11.52.



Ordering a Confidence Interval

Only one of these sentences is defensible, and the wrong one sounds better.

The Sentence to Avoid

“There is a 95 per cent chance the true average lies between 11.19 and 13.31.”

The Sentence to Use

“Our estimate is 12.25 euro, with a 95 per cent confidence interval from 11.19 to 13.31.”

The first sentence makes μ vary; only the endpoints vary (Hoekstra et al., 2014).

Testing the Mean Against a Claimed Value

Somebody claims average hourly earnings in this population are ten euro. That claim is the null hypothesis, and the statistic below is the t-statistic:

$$t = \frac{\bar{y} - \mu_0}{\text{se}(\bar{y})} \quad (4)$$

- μ_0 : the null hypothesis, the value the claim asserts, here 10.00
- t : how many standard errors $\bar{y}$ sits from μ_0

t Equals 4.2, Far Into the Tail

$(12.25 - 10.00)/0.54 = 4.2$, far into the tail of what the claim would produce.

What a p-Value Says and What It Does Not

A p-value says what the claim would produce, not whether it is true. It measures how badly the data embarrasses the claim, not its truth:

- **What a p-Value Is:** The smallest error rate at which the claim would be rejected.
- **Under the Claim:** Fewer than one sample in ten thousand reaches $|t| = 4.2$.

Not the Chance of Error

A p-value is not the probability the claim is false. Nor is a p-value the chance we erred. A large p-value is no evidence the claim is true (Wasserstein & Lazar, 2016).

Significance and the Size of the Gap

A large statistic comes from a large gap or a small standard error:

- **Statistical Significance:** The gap divided by its standard error, which is t .
- **Practical Significance:** The size of the gap itself, here 2.25 euro an hour.
- **Why Both Are Reported:** A big sample makes a tiny gap statistically significant.

Significance Is Not Magnitude

Whether 2.25 euro an hour matters is a judgement, not a test result.

What Would Have To Be True?

Question

What would have to be true for this number to mean what we just said it means? And what is the plausible story where it isn't?

The 95 per cent belongs to the procedure, and the procedure includes sampling. If the 120 chose themselves, the rule covers less than 95 per cent. The arithmetic on the slide will not say so.

Application

Simulate the whole exercise, and count:

- **Draw:** 120 people, take the mean, build the interval by the same rule.
- **Repeat:** Five hundred times, recording whether 11.52 falls inside.
- **Change:** The sample size, then see which number shrinks and which holds.

Coverage Holds, Width Shrinks

Raising n narrows every interval and leaves coverage near 95 per cent.

Three Claims About an Estimate

Today was three separate ideas, each doing one job:

1 What Makes an Estimator Good.

Unbiased, consistent and efficient are three separate properties.

2 The Sampling Distribution.

How an estimate varies across samples nobody drew.

3 Confidence Intervals and Tests.

Two claims about the procedure, not about the number.

Before Lecture 2.1

1 Play:

The sampling app, at $n = 30$ and then $n = 480$.

2 Apply:

Problem set 1, covering 1.2 and 1.3.

3 Review:

Wooldridge (2020), Math Refresher C, on estimation and inference.

Problem Set 1 This Week

[Out this week, due week n . The last question is the coverage simulation, written as prose not code.]

Next Week

Lecture 2.1 stops averaging one variable and fits a line through two:

- **The Population Line:** An intercept, a slope on schooling, and an error.
- **The Least Squares Rule:** One rule turning 120 pairs into a slope.
- **The Same Three Questions:** Unbiased, how variable, and what interval.

The Slope Is an Estimator

Everything today said about $\bar{y}$ gets asked again of the fitted slope.

Thank you

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BACKUP

Appendix

Derivation: Unbiasedness and Consistency of the Sample Mean

Take the expectation of the sample mean. Then take its variance and grow n .

Linearity, Then Independence

Linearity of expectation carries part one. Independence of the draws carries part two, which is what random sampling buys.

[Full derivation. Set as homework; worked solution published with the set.]

Derivation: The Standard Error of the Mean

Compute the variance of the sample mean, then take its square root. Replace the population spread with the sample one.

Root n Is Not Assumed

The root n appears only at the last step, from taking the square root.

[Full derivation. Set as homework; worked solution published with the set.]

Notes and Tools

- The app, and this week's simulation script.
- `mean()`, `sd()` and `replicate()` in R, which are the whole of the lab.
- Simulation as the route in (Chance et al., 2004; delMas et al., 1999; Tintle et al., 2012).
- The two misreadings this lecture heads off (Greenland et al., 2016; Kalinowski et al., 2008; Sotos et al., 2007).
- Retrieval practice, and why it beats restudy (Crouch & Mazur, 2001; Roediger & Karpicke, 2006).
- What a first course should spend its time on (Angrist & Pischke, 2017).

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